

KALP SHAH

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Education

Indian Institute of Technology Gandhinagar

Bachelor of Technology in Electrical Engineering

Jul. 2024 – Apr. 2028

Gandhinagar, India

Experience

AIResQ ClimSols

Machine Learning Intern

May 2026 – July 2026

Gandhinagar, India

- Engineered a **Computer Vision** pipeline for estimating depth of floods using CCTV Camera Networks
- Developed a **custom Camera and Radar device** for generation and validation of data in flood areas
- Implemented **single view metrology** algorithms for extracting features achieving **accuracy of $\pm 3\text{cm}$**

Barclays BIIC Mentorship Program

Mentee

June 2026 – July 2026

- Selected as **1 of 30 students** across top IITs for Barclays' mentorship program focused on Capital Markets
- Built an FX strategy backtester; ranked in the **top 5** participants selected to present the strategy to the Barclays team

ABSix Robotics

Software Development Intern

May 2025 – July 2026

Gandhinagar, India

- Engineered a **cross-platform desktop application (Python, Qt)** with real-time robot-student communication via the **Dynamixel SDK** deployed across **15+ challenges** and actively used by **100+ students**
- Integrated a **RAG pipeline (LangChain + ChromaDB)** into the platform to query robotics concepts

Projects

Belief Tracking for Countering LLM Negotiation Agents using Hidden Markov Models

Supervisor: Prof. Manisha Padala

Github

- Modeled LLM negotiation behavior** through a bargaining game simulation as a time-varying latent-state process using Hidden Markov Models
- Implemented HMM inference** using Forward-Backward, Viterbi, and Baum-Welch algorithms to track posterior beliefs and recover the opponent's hidden negotiation regime over time
- Developed belief-driven negotiation agents** and benchmarked them against no-belief, frequentist, and static-Bayesian baselines using negotiation payoff, regime-detection accuracy, and detection-lag metrics

Realtime Carbon Credit Market Analysis Platform

Inter-IIT Technical Meet 14.0 – December 2025

Github

- Architected a **full-stack real-time analytics platform** (Next.js, WebSocket, PostgreSQL) serving live carbon credit pricing, trend signals, and risk analysis to end users as part of an 8-person team
- Developed a **multi-agent RAG system** (LangChain, DeepSeek) generating context-aware recommendations
- Built a **data streaming pipeline** using **Kafka** and **Debezium** (CDC) to capture live **PostgreSQL** changes, cutting data freshness latency from **60 minutes to under 3 seconds**

Controllable Font Generation using cGANs

Github

- Engineered a conditional dataset from **281,000+ TMNIST images**, enabling precise multi-label typographic control by parsing implicit font nomenclature into explicit binary style vectors
- Architected a **Conditional Generative Adversarial Network**, modeling a 784-dimensional feature space
- Synthesized 94-character font families, ensuring consistent stylistic features across the generated alphabet by locking latent noise vectors during inference achieving an **FID Score of 25.7466 over 1000 generations**

Technical Skills

Languages: Python, C++, C, HTML/CSS, JavaScript, SQL

Frameworks: PyTorch, OpenCV, Langchain, scikit-learn, Selenium, Next.js

Cloud/DevOps: Azure, Linux, Docker, Git, PostgreSQL

Achievements

Codeforces Rating – Handle: *kalpshah1808* | Expert: 1743

Finalist at AMS Derive 2026 conducted by Jane Street and Qube Research at IIT Bombay

JEE Advanced 2024 – All India Rank 6637 and JEE Mains 2024 – All India Rank 3917

Integral Cup Season 1 Prelims All India Rank 47 out of 2500+ students conducted by Optiver

Awarded a Department Change from B.Tech-M.Tech Mechanical to Electrical Engineering after First Year